



Run It Top to Bottom

Quick facilitation notes & answer key

Goal

Children learn that a block program runs from the top block down, one block at a time, and that the position of a block decides when it runs. This builds the core idea of sequential execution: the order of the blocks IS the algorithm.

Kindergarten frame (Ontario)

Problem Solving and Innovating — children determine and describe the order in which a set of instructions is carried out.

No reading required of the child: read each prompt aloud; the child circles a block and writes order numbers (1-4).

Materials

The printed worksheet and a pencil. Optional: the 4 paper blocks from Sheet 1 plus an up ↑ and left ← block to physically point to in order.

Run it (about 15 min)

1. MODEL: point to Bit's stack and tap each block in turn — '■ runs first... ➡ second... ↑ third... ← last.' One tap per block, top to bottom.
2. GUIDED: Exercise 1 — circle the top ■ block (runs first). Exercise 2 — children write 1, 2, 3, 4 down the stack.
3. YOU TRY: Exercise 3 — on the second stack, circle the bottom ➡ (runs last) and number 1, 2, 3.
4. Connect: 'A computer runs its blocks in order, top to bottom — it never skips around.'

Answer key (modelled by the run-gate)

Ex 1 — Bit runs the top ■ GO block first.

Ex 2 — order down the stack: ■ = 1, ➡ = 2, ↑ = 3, ← = 4.

Ex 3 — the last block is the bottom ➡; order down the second stack: ■ = 1, ← = 2, ➡ = 3.

Watch for & differentiate

Common slip: numbering bottom-to-top. Re-anchor: 'We always start at the TOP, like reading starts at the top of the page.'

Simplify: cover all blocks but the top, reveal one at a time as you count.

Extend: ask 'if we swapped the ➡ and the ←, would Bit do the same thing?' (No — order matters.)

Movement option: stand 4 children in a line holding blocks; tap shoulders top-to-bottom so each acts in turn.

Success looks like

The child numbers the blocks 1, 2, 3, 4 from the top down and names the first and last blocks correctly for both stacks.